SC- and PC-NEVPT2 and QD-NEVPT2 in free-breakfast

Energies, analytic nuclear gradients and nonadiabatic couplings, density-fitted, on the GPU

free-breakfast implementation notes

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Abstract

This document describes the `src/nevpt2` layer of free-breakfast: the strongly contracted (SC) and partially contracted (PC) n-electron valence state perturbation theory (NEVPT2) energies of a CASSCF reference, their quasidegenerate (QD) effective Hamiltonians over a state-averaged CASSCF, the analytic nuclear gradient of every one of them, the QD interstate coupling and derivative (nonadiabatic) coupling, and what a QD minimum-energy conical intersection search needs. Everything is density fitted and runs on the device. The organising idea is that every correlation quantity is a function of only three objects, the MO one-electron matrix h, the fitted three-index tensor B and the CI vectors, so one reverse pass through the energy gives its adjoints $(\bar{h}, \Lambda, \bar{y})$, and from there each derivative is the CASSCF machinery the tree already has: one Z-vector solve over the CI-relaxed Hessian and one or two passes of the density-fitted derivative integrals. No derivative is written by hand beyond a few scalar finishes, and no finite difference appears in any production path.

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1 Scope and layering

What is implemented.

- SC- and PC-NEVPT2 class energies of one root [1, 2, 3], over a Dyall zeroth-order Hamiltonian [5].
- SC- and PC-QD-NEVPT2 [4]: the effective Hamiltonian of n roots of a state-averaged CASSCF, in the "SA-NEVPT2" scheme of Park [10] (one set of orbitals, core and virtuals canonical over the state-averaged Fock), symmetrised and diagonalised.
- Analytic nuclear gradients of single-state SC and PC-NEVPT2 on a single-state DF-CASSCF [8, 9], and of any SC- or PC-QD-NEVPT2 state on a DF-SA-CASSCF at any weights [10].
- The QD interstate coupling $\mathbf{h}_{PQ}$, and the full derivative coupling $\mathbf{d}_{PQ} = \langle \Psi_P | \nabla \Psi_Q \rangle$, including the PT2 term [10, 11].
- For conical-intersection searches: $\mathbf{g}_P$, $\mathbf{g}_Q$ and $\mathbf{h}_{PQ}$ at one set of H^{eff} eigenvectors with no refusal of a degenerate pair, and bounds on how far the pair's gap moves when the SA-CASSCF is not fully converged.

Layering. `src/nevpt2` sits over `core`, `integrals` (the density-fitted MO transform), `ci` (the strings, sigma vectors and RDMs) and, for derivatives, `response` and `mcscf` (the CASSCF Fock, orbital Hessian, Z-vector, DF gradient and CASSCF couplings). It never imports `pipeline`; the pipeline's `[nevpt2]`, `[nevpt2_gradient]`, `[nevpt2_nac]` and `[optimize]` steps wrap it. Its six sub-modules are:

module	role
`nevpt2.density_fitting`	MO blocks of B, the active $(tu\|vw)$, the ten external blocks slab by slab, and their adjoints onto Λ (§4.1)
`nevpt2.rdm`	dm3 and the integral-contracted 4-RDMs f3ca, f3ac, of one vector or between two, and their adjoints (§4.2)
`nevpt2.energy`	the eight SC and PC class energies, the SC- and PC-QD effective Hamiltonians, the amplitude hand-over, and the reverse passes of all four (§§4–5)
`nevpt2.gradient`	the energy as a function of (h, B, C, c), z^c, the Lagrangians, Z-vectors and nuclear terms; single-state and QD, SC and PC (§§6–7)
`nevpt2.nac`	the PT2 transition density and the QD derivative coupling (§8)
`nevpt2.intersection`	a QD pair's branching space and gap sensitivities (§9)

Provenance. The energy kernels are a port of the nevpt2-gpu-demo, itself a transcription of PySCF's `mrpt.nevpt2` [16]: every einsum keeps the reference's subscripts, so the two read side by side. The PC energy is validated against block2's `WickICNEVPT2` [17]. The derivative layers are this tree's own. A related device-resident approach, DMRG-QD-SC-NEVPT2 gradients and couplings as one reverse-mode transpose of a contraction graph, is Ref. [18].

2 Notation

Orbitals. C ($n_{\mathrm{ao}} \times n_{\mathrm{mo}}$) maps AOs μ, ν to MOs, ordered core, active, virtual. Core (inactive) labels are i, j, k, l, active t, u, v, w (and p, q, r, s inside active-only tensors), virtual a, b (and r, s as the external labels of a class), general x, y, z. Counts are n_c, n_a, n_v, n_{mo}. A rotation is $C \to Ce^{\kappa}$, κ real antisymmetric.

Integrals. $h = C^{\mathsf{T}} h^{\mathrm{AO}} C$. With three-centre integrals $(\mu\nu|P)$ and the Coulomb metric $(P|Q) = LL^{\mathsf{T}}$,

$$B^L_{xy} = \sum_P (L^{-1})_{LP} \left(C^{\mathsf{T}}(\cdot\,\cdot\,|P)\, C\right)_{xy}, \qquad (xy|zw) = \sum_L B^L_{xy} B^L_{zw}, \tag{1}$$

the density-fitting approximation [15]. Every two-electron quantity below is built from B through such products.

Densities. Spin-summed E_{xy}; active RDMs $\Gamma^{(1)}_{tu} = \langle E_{tu}\rangle$, dm2 $= \langle E_{tu}E_{vw}\rangle$, dm3 $= \langle E_{tu}E_{vw}E_{xy}\rangle$, in PySCF's unreordered layout. Between a bra $\langle M|$ and a ket $|N\rangle$, $\gamma^{(MN)}_{tu} = \langle M|E_{tu}|N\rangle$ and so on, unsymmetrised.

Fock matrices. The inactive Fock and the generalised Fock are

$$F^I_{xy} = h^{\mathrm{eff}}_{xy} = h_{xy} + \sum_j \big[2(xy|jj) - (xj|jy)\big], \qquad f_{xy} = F^I_{xy} + \sum_{tu} \gamma_{tu}\big[(xy|tu) - \tfrac{1}{2}(xt|uy)\big], \tag{2}$$

and the orbital energies are $\varepsilon_x = f_{xx}$ for x core or virtual. On a state average $\gamma \to \gamma^{\mathrm{SA}} = \sum_r w_r \Gamma^{(1)}_r$.

3 NEVPT2 and QD-NEVPT2

3.1 The zeroth-order Hamiltonian and the classes

NEVPT2 partitions $H = H^{(0)} + V$ with Dyall's Hamiltonian [5]

$$H^D = \sum_i \varepsilon_i E_{ii} + \sum_a \varepsilon_a E_{aa} + \sum_{tu} h^{\text{eff}}_{tu} E_{tu} + \tfrac{1}{2} \sum_{tuvw} (tu|vw)\big(E_{tu}E_{vw} - \delta_{uv}E_{tw}\big) + C, \tag{3}$$

and splits the perturbation into eight classes $V_l^{(k)}$ by how many electrons the active space gains (k) and from which external orbitals (l) [1]. Their PySCF names are used throughout:

name	space	external tuple	Δ (orbital-energy part of the denominator)
Sr	$S_r^{(-1)}$	a	ε_a
Si	$S_i^{(+1)}$	i	$-\varepsilon_i$
Sijrs	$S_{ijab}^{(0)}$	(i,j,a,b)	$\varepsilon_a + \varepsilon_b - \varepsilon_i - \varepsilon_j$
Sijr	$S_{ija}^{(+1)}$	(i,j,a)	$\varepsilon_a - \varepsilon_i - \varepsilon_j$
Srsi	$S_{abi}^{(-1)}$	(a,b,i)	$\varepsilon_a + \varepsilon_b - \varepsilon_i$
Srs	$S_{ab}^{(-2)}$	(a,b)	$\varepsilon_a + \varepsilon_b$
Sij	$S_{ij}^{(+2)}$	(i,j)	$-\varepsilon_i - \varepsilon_j$
Sir	$S_{ia}^{(0)}$	(i,a)	$\varepsilon_a - \varepsilon_i$

Every class but Sijrs has active indices; Sijrs reads no RDM.

3.2 The SC energy

In SC-NEVPT2 each external tuple t has one perturber, $P_tV|\Psi_0\rangle$. Its norm $N_t = \langle\Psi_0|V^\dagger P_t V|\Psi_0\rangle$ and its Dyall element H_t (the active part of $\langle P_tV\Psi_0|H^D - E_0|P_tV\Psi_0\rangle$) are quadratic forms of a slab row of integrals over RDM-built matrices, and

$$E^{\text{SC}} = \sum_k \sum_{t\in k} E_t, \qquad E_t = -\frac{N_t}{\Delta_t + H_t/N_t}, \tag{4}$$

over the perturbers with $|N_t| > 10^{-14}$ (PySCF's `_norm_to_energy` guard; Sijrs has none). The matrices are the metrics (dm3, dm2, dm1 and their hole forms) and the Dyall blocks `a16`, `a22`, ... of PySCF, which for Sr and Si read the 4-RDM only through f3ca and f3ac (§4.2).

3.3 The PC energy

PC-NEVPT2 keeps the span of each class's internal functions [3]. For a class with metric S, Dyall matrix K, one-body metric S_1 and a coupling factor c, a tuple's slab row x_t and its one-body row h_t give

$$E_t = -c\, b_t^{\mathsf{T}} (K + \Delta_t S)^{+} b_t, \qquad b_t = S x_t + S_1 h_t, \tag{5}$$

$(\cdot)^+$ the inverse on the span of S that the metric cut keeps. S, K and S_1 are the same matrices SC builds (SC contracts each against one row; PC solves with them), so PC is "SC without the final contraction":

class	basis, d	S	K	S_1	c
Sr	(p,q,r), n_a^3	dm3$[r,p,q,b,a,c]$	a16	dm2$[r,p,q,a]$	1
Si	(p,q,r), n_a^3	dm3_h$[r,p,q,b,a,c]$	a22	dm2_h$[r,p,q,a]$	1
Sijr	p, n_a	hdm1	a3	—	$2\,\mathrm{direct}-\mathrm{exchange}$
Srsi	p, n_a	dm1	k27	—	$2\,\mathrm{direct}-\mathrm{exchange}$
Srs	(p,q), n_a^2	rm2$[p,q,b,a]$	a7	—	$\frac{1}{2}$
Sij	(p,q), n_a^2	hdm2	a9	—	$\frac{1}{2}$
Sir	$(p,q)\oplus(p,q)$, $2n_a^2$	spin-coupling blocks	a12, a13	$[2\,\mathrm{dm1};\,-\mathrm{dm1}]$	1
Sijrs	none	—	—	—	SC's

3.4 QD-NEVPT2

Over n roots $|M\rangle$ of a state-averaged CASSCF sharing one set of orbitals, with ε from f^{SA} for every root (Zaitsevskii–Malrieu multipartitioning [6], Park's SA-NEVPT2 scheme [10]), each root has its own perturber space and Dyall Hamiltonian, and the couplings are $C_{MN} = \langle M|V|\Psi_N^{(1)}\rangle$. With $\Psi_N^{(1)}$ the ket's first-order function:

$$\text{SC:}\quad C_{MN} = -\sum_t \frac{\bar{T}^t_{MN}}{\Delta_t + H_t^N/N_t^N}, \qquad \bar{T}^t_{MN} = \tfrac{1}{2}(T^t_{MN} + T^t_{NM}), \tag{6}$$

$$\text{PC:}\quad C_{MN} = -c\sum_t b_t^{M\leftarrow N\,\mathsf{T}}\left(K_N + \Delta_t S_N\right)^{+} b_t^N, \qquad b_t^{M\leftarrow N} = S^{(N,M)}x_t + S_1^{(N,M)}h_t. \tag{7}$$

T^t_{MN} is the class's norm einsum over pair (M,N)'s transition RDMs; the terms of it that read no RDM carry $\langle M|N\rangle = 0$ and are dropped. The mean over (M,N) and (N,M) in SC undoes PySCF's factor 2 on the Hermitian cross terms of Sr, Si and Sir, which holds only for a state's own RDMs. $S^{(N,M)}$, $S_1^{(N,M)}$ are the PC metrics over pair (N,M)'s transition RDMs (bra N, ket M); S_1 must be that orientation. PC has no mean, so its C is not symmetric. Sijrs is diagonal. The effective Hamiltonian is symmetrised as in MS-CASPT2 [10]:

$$H^{\mathrm{eff}}_{MN} = E_M\delta_{MN} + \tfrac{1}{2}(C_{MN} + C_{NM}), \qquad H^{\mathrm{eff}}R_P = E_P R_P, \quad R^{\mathsf{T}}R = 1, \tag{8}$$

each column of R signed so its largest-magnitude component is positive.

4 The energy on the device

4.1 Density-fitted integrals, slab by slab

`mo_blocks` forms the five MO blocks B_{aa}, B_{ca}, B_{va}, B_{cv} and B_{vc} (B_{cv} with its pair swapped, so every batched index is the first of its pair). No $(Q|vv)$ block exists: no class has two virtuals in one charge distribution. The ten external blocks are built a slab at a time along their batch axis, so no whole external block is ever on the device:

$$\mathcal{C}_{xyzw} = (xy|zw) = \sum_Q X_{Q,xy}\,Y_{Q,zw}, \qquad \text{slab} = \mathrm{permute}(\mathcal{C}), \tag{9}$$

one strided-batched GEMM over y ($\mathcal{C}_y = X_y^{\mathsf{T}}Y$) and one permute kernel per slab; Sijrs needs no permute.

block	physicist slab	X	Y	axes	batch
Sr	$[r,a,b,c]=(rb\|ac)$	B_{va}	B_{aa}	0213	0
Si	$[t,u,i,v]=(it\|uv)$	B_{ca}	B_{aa}	1203	2
Sijrs	$[i,a,j,b]=(ia\|jb)$	B_{cv}	B_{cv}	0123	0
Sijr	$[r,p,j,i]=(rj\|ip)$	B_{vc}	B_{ca}	0312	0
Srsi	$[x,y,i,t]=(xi\|yt)$	B_{vc}	B_{va}	0213	0
SrsiT	$[y,x,i,t]=(xt\|yi)$	B_{va}	B_{vc}	2031	1
Srs	$[r,s,t,u]=(rt\|su)$	B_{va}	B_{va}	0213	0
Sij	$[t,u,i,j]=(it\|ju)$	B_{ca}	B_{ca}	1302	2
Sir1	$[r,t,i,u]=(ri\|tu)$	B_{vc}	B_{aa}	0213	0
Sir2	$[r,u,t,i]=(rt\|iu)$	B_{va}	B_{ca}	0312	0

Adjoints. With $\bar{\mathcal{C}}$ the slab bar permuted back, $\bar{X}_y \mathrel{+}= Y\bar{\mathcal{C}}_y^{\mathsf{T}}$ and $\bar{Y}_w \mathrel{+}= X_b\bar{\mathcal{C}}_w$, two strided-batched GEMMs written straight into $\Lambda = \partial E/\partial B$ ($n_{\text{aux}} \times n_{\text{mo}}^2$, B's own layout) at each block's corner, so no block bar is ever formed. A slab's bar is consumed by its call and freed before the next slab is built.

4.2 RDMs: dm3 and the contracted 4-RDMs

The 4-RDM never exists. Sr and Si read it only contracted with $(tu|vw)$:

$$\text{f3ca}_{pqrstu} = \sum_{ijk}(ki|uj)\,\langle E_{pq}E_{rs}E_{tj}E_{ki}\rangle, \qquad \text{f3ac}_{pqrstu} = \sum_{ijk}(ik|jt)\,\langle E_{pq}E_{rs}E_{ju}E_{ik}\rangle, \tag{10}$$

(here i,j,k active). `nevpt2::rdms` builds dm3, f3ca and f3ac (n_a^6 each), tiled over determinants K against a byte budget. Per tile, one thread per determinant gathers

$$R_{tu,K} = \langle K|E_{tu}|c\rangle, \qquad L2_{pqrs,K} = \langle c|E_{pq}E_{rs}|K\rangle \tag{11}$$

from the strings' own replacement lists (Knowles–Handy [14]), then dm3 $\mathrel{+}= L2\,R^{\mathsf{T}}$ is one GEMM; per order the consume $W = (tu|vw)\cdot L2$ is n_a strided-batched GEMMs and $f3 \mathrel{+}= L2\,W^{\mathsf{T}}$ one GEMM. After the last tile a contraction of $(tu|vw)$ with dm3 (fdm2) and PySCF's wedge reconstruction (the commutator $E_{pq}E_{rs} = E_{rs}E_{pq} + \delta_{qr}E_{ps} - \delta_{ps}E_{rq}$) run once per order. Nothing is atomic, so the result does not depend on the run. The cost is $O(N_{\text{det}}n_a^8)$ in GEMMs, the same order as forming the 4-RDM, with $O(N_{\text{det}}n_a^4)$ memory per tile.

Between two vectors. The bra/ket build gathers $L2$ over the bra and R over the ket, so $L2\,R^{\mathsf{T}} = \langle M|EEE|N\rangle$; dm1 and dm2 come from one GEMV each per tile. The consume reads the ket's own $L2$ (a second gather), and the ca order's fdm2 reads the transition element that hermiticity let the one-vector build skip. With `bra == ket` it is the one-vector build operation for operation. QD reads f3 on the diagonal only, so the pair loop passes null f3 pointers and only dm1–dm3 are built, about a third of the GEMM work.

Adjoints. `rdms_adjoint` runs the build backwards over the same tiles: the wedge and fdm2 first, then per tile each digest acc $\mathrel{+}= L2\,B^{\mathsf{T}}$ as two GEMMs ($\bar{L2} \mathrel{+}= \overline{\text{acc}}\,B$, $\bar{B} = \overline{\text{acc}}^{\mathsf{T}}L2$), each consume as two loops of batched GEMMs, and one gather onto the CI bar over each determinant's replacement lists. dm1 and dm2 have no bars of their own: since $\sum_t E_{tt} = N$ on the ket,

$$\text{dm2}_{pqrs} = \tfrac{1}{N}\sum_t \text{dm3}_{pqrstt}, \qquad \text{dm1}_{pq} = \tfrac{1}{N^2}\sum_{rt}\text{dm3}_{pqrrtt}, \tag{12}$$

so their bars fold into $\overline{\text{dm3}}$ along those traces, and one adjoint carries every RDM. Neither a sigma vector nor any symmetry of a bar is assumed.

4.3 The SC and PC class walks

Each class builds its RDM-only intermediates once, then walks its external index in batches: the slab is built, contracted by the class's einsums (`recipes::einsum`, with the reference's subscripts unchanged) into per-perturber N_t and H_t, read back and finished on the host, Eq. (4). Srsi pairs (r, s) with (s, r) across slabs, so it fills its whole (v, v, c) tables first.

PC solves each class once, since S, K and S_1 do not depend on the tuple (Algorithm 1). Each slab is then one einsum into the eigenbasis.

Algorithm 1 A PC class (`solve_class`, then the walk)

Input: S, K, S_1 ($d \times d$, $d \times n_1$) of one state; the class's slabs

1: $S \leftarrow \operatorname{sym} S$, $K \leftarrow \operatorname{sym} K$ ▷ K is symmetric only to the CI's convergence
2: $S = V\sigma V^{\mathsf{T}}$; keep $\sigma_k > 10^{-13}\sigma_{\max}$
3: **gap rule:** $\sigma_{\min,\text{kept}} \geq 10^3\,\nu$, ν the largest dropped σ, or $\varepsilon_{\text{mach}}\sqrt{d}\,\sigma_{\max}$ if none is dropped; else refuse
4: $X \leftarrow V_{\text{kept}}\sigma_{\text{kept}}^{-1/2}$; $K' = X^{\mathsf{T}}KX = W\lambda W^{\mathsf{T}}$; $U \leftarrow XW$
5: $\mathcal{T} \leftarrow SU$; $t_1 \leftarrow S_1^{\mathsf{T}}U$
6: **for** each slab, each tuple t **do**
7: $\quad y_t \leftarrow \mathcal{T}^{\mathsf{T}}x_t + t_1^{\mathsf{T}}h_t$ ▷ one einsum per slab
8: $\quad$ refuse if $\min_k(\lambda_k + \Delta_t) \leq 0$
9: $\quad E_t \leftarrow -c\sum_k y_{tk}^2/(\lambda_k + \Delta_t)$

S is singular by construction (Sr and Si drop 44 of 64 modes at CAS(4,4)). The gap rule refuses a cut so near the threshold that round-off decides the rank, and so the energy. A refused class fails the call (naming the class, and for QD the state); it is never summed in as zero.

4.4 The QD walks and the amplitude hand-over

`sc_effective_hamiltonian` and `pc_effective_hamiltonian` take $n \times n$ pair inputs, entry (M, N) carrying the shared integrals and pair (M, N)'s transition dm1–dm3 (f3 on the diagonal only). Each class walks its slabs once for every pair. For SC, per slab, one table per given pair accumulates T^t_{MN} and one per given state N^N_t, H^N_t; the Dyall blocks are built from each state's own RDMs only. For PC, each ket N's $(S_N, K_N, S_{1,N})$ is solved once, the transitions' $\mathcal{T}_{M\leftarrow N} = S^{(N,M)\mathsf{T}}U_N$ and $t_{1,M\leftarrow N} = S_1^{(N,M)\mathsf{T}}U_N$ are built once per pair, and

$$C^t_{MN} = -c\sum_k \frac{y_{M\leftarrow N,tk}\; y_{N,tk}}{\lambda_{N,k} + \Delta_t}, \qquad y_{M\leftarrow N,t} = \mathcal{T}^{\mathsf{T}}_{M\leftarrow N}x_t + t^{\mathsf{T}}_{1,M\leftarrow N}h_t, \tag{13}$$

with Sijr and Srsi's "2 direct − exchange" as the bilinear $\sum_t(2y_{M\leftarrow N,t}y_{N,t} - y_{M\leftarrow N,t}y_{N,t'})/(\lambda + \Delta_t)$.

The hand-over. Both calls can return each state's first-order amplitudes: for SC $c_t = -1/(\Delta_t + H^N_t/N^N_t)$ (zero where the guard drops the perturber), so $\Psi^{(1)}_N \ni c_t P_t V|N\rangle$; for PC U_N and $g_{tk} = y_{N,tk}/(\lambda_{N,k} + \Delta_t)$, so $\Psi^{(1)}_N \ni -\sum_k g_{tk}\sum_\mu U_N[\mu, k]\phi^N_\mu$. A later call with its diagonal omitted reads them instead of walking the states again: n states take one diagonal call, then one call per pair over (M, N) and (N, M) alone, holding two transition sets at a time and never n^2. The PT2 transition density (§8.3) contracts the same amplitudes.

5 The reverse passes

5.1 One derivative rule

`einsum_adjoint` takes a forward einsum's subscripts, operands, output shape and scale unchanged, and writes operand k's bar as the same einsum with the output and operand k swapping roles. An operand that repeats a label takes its bar over its distinct labels and is scattered onto the diagonal; a strided view takes its bar packed and one GEMM-add places it. Every class lists its norm and H terms once, as data, and both directions read the list, so no derivative of a class is transcribed by hand. The NumPy slice assignments of the reference (`a16[:, i, :, :, :, i] += fdm2` and two in a22) have the gather fdm2^{T}-bar = `einsum("pirabi->prab", a16_bar)` as their adjoint, since an input label may repeat.

5.2 The SC finish

The only hand-differentiated expression on the SC side is Eq. (4)'s finish. With $T_t = -1/(\Delta_t + H_t/N_t)$ [8]:

$$P_t = \frac{\partial E}{\partial N_t} = 2T_t + \Delta_t T_t^2, \qquad Q_t = \frac{\partial E}{\partial H_t} = T_t^2, \qquad R_t = \frac{\partial E}{\partial \Delta_t} = N_t T_t^2, \tag{14}$$

zero for a dropped perturber. P and Q go back through the norm and H einsums onto the slabs (and so Λ), the one-body rows and the blocks' bars $M_G = \partial E/\partial G$; R goes onto the orbital energies Δ_t reads. Sijrs has its own kernel. Srsi walks twice, since its finish pairs entries across slabs.

The sweep. The block bars are swept back through the builders that formed them, in reverse topological order (an intermediate's bar is complete before it is swept), onto $\bar{h}$, the active $\overline{(tu|vw)}$ and $\overline{\mathrm{dm1}}, \ldots, \overline{\mathrm{f3ac}}$. It has two halves: the Dyall blocks, which read h and $(tu|vw)$ and only a state's own walk has, and the metrics, which read only RDMs and are all a QD transition takes. Intermediates are recomputed, never kept.

Identities from the bars. Each class energy is homogeneous of degree one in (N, H), so

$$\sum_t (P_t N_t + Q_t H_t) = \sum_G \langle M_G, G\rangle = E_k \tag{15}$$

(Park's Eq. 40 [8]); and at fixed orbital energies E is homogeneous of degree 4 in $(B, h_v^{1/2})$, so $\langle \Lambda, B\rangle + 2\sum \langle \bar{h}_v, h_v\rangle = 4E$, h_v the one-body rows. Both are step-free checks of the bars.

5.3 The PC solve, backwards

With $A_t = K + \Delta_t S$ and $x = A_t^+ b_t$ the solve the energy ran,

$$\dot{E}_t = -c(2\, x^{\mathsf{T}} \dot{b}_t - x^{\mathsf{T}} \dot{A}_t\, x), \tag{16}$$

so no eigenvalue or eigenvector derivative appears. This holds where S's kept rank is locally constant (the gap rule) and K annihilates S's dropped modes, which holds at an eigenstate. In the eigenbasis, with $g_t = y_t/(\lambda + \Delta_t)$, each tuple gives $\bar{y}_t = -2c\, g_t$, a weight $w_t = c\, g_t$ and $\partial E/\partial \Delta_t = c\, g_t \cdot g_t$; accumulating $M = \sum_t g_t w_t^{\mathsf{T}}$ and $M_\Delta = \sum_t \Delta_t g_t w_t^{\mathsf{T}}$ over the walk,

$$\frac{\partial E}{\partial K} = U M U^{\mathsf{T}}, \qquad \frac{\partial E}{\partial S} = U \bar{T}^{\mathsf{T}} + U M_\Delta U^{\mathsf{T}}, \qquad \frac{\partial E}{\partial S_1} = U \bar{t}_1^{\mathsf{T}}, \tag{17}$$

symmetrised as the solve symmetrised. Those go back through the einsums that built S, K, S_1 onto SC's block bars, and SC's sweep takes them unchanged. At fixed slabs and orbital energies a PC class is homogeneous of degree one in (S, S_1, K), which gives PC's analogue of Eq. (15).

5.4 The QD reverse passes

Both QD passes differentiate $L = \sum_{MN} \bar{H}_{MN} H_{MN}$ for a given $\bar{H}$; at $n = 1$, $\bar{H} = 1$ they are the single-state passes einsum for einsum.

SC-QD. With $D_N = \Delta + H^N/N^N$, the coupling $-\bar{T}_{MN}/D_N$ has three bars:

$$\frac{\partial L}{\partial \bar{T}_{MN}} = -\frac{\bar{H}_{MN}}{D_N}, \qquad A_N = \sum_M \frac{\bar{H}_{MN}\bar{T}_{MN}}{D_N^2}, \qquad (\bar{N}^N, \bar{H}^N, \bar{\Delta}) \mathrel{+}= \Big(-\frac{A_N H^N}{(N^N)^2},\ \frac{A_N}{N^N},\ A_N\Big), \quad (18)$$

the first split half and half onto the norm einsums of (M, N) and (N, M). The walk runs in three parts: each state's tables, evaluated only; each unordered pair, its norm einsums evaluated and differentiated at once, its metric bars swept onto its transition dm1–dm3 and handed to a callback, then freed; each state's own walk with A_N added. Memory is one pair's transition set at a time, never n^2 sets.

PC-QD. The coupling is bilinear in the two rows, so with $g_M = y_{M\leftarrow N}/(\lambda_N + \Delta)$ and $g = y_N/(\lambda_N + \Delta)$ the bars are $-c\bar{H}_{MN} g$ on $y_{M\leftarrow N}$, $-c\bar{H}_{MN} g_M$ on y_N, $c\bar{H}_{MN} g_M g^{\mathsf{T}}$ on M_N (and Δ times it on $M_{\Delta,N}$), and $c\bar{H}_{MN} g_M \cdot g$ on Δ. Still no eigenvector derivative: $b^{M\leftarrow N}$ lies in S_N's kept range, since a mode annihilating $|N\rangle$ annihilates $\langle N|S^{(N,M)}$. Then $\partial L/\partial S^{(N,M)} = \bar{\mathcal{T}}_{M\leftarrow N} U_N^{\mathsf{T}}$ and $\partial L/\partial S_1^{(N,M)} = U_N \bar{t}_{1,M\leftarrow N}^{\mathsf{T}}$, unsymmetrised, onto pair (N, M)'s transition RDMs. Each class walks its slabs once with every pair, as the forward does.

6 The single-state gradient

6.1 The energy as a function of (h, B, C, c)

`sc_energy` and `pc_energy` rebuild everything the class walk reads from (h, B) in the MO basis of C and the CI vector c:

$$h1e = F^I_{tu}, \quad h1e_v_si = F^I_{ti}, \quad h1e_v_sir = F^I_{ai}, \quad h1e_v_sr = F^I_{at} - \sum_u (au|ut), \quad \varepsilon_x = f_{xx}, \tag{19}$$

dm3, f3ca, f3ac by `rdms`, and dm1, dm2 as dm3's traces (Eq. (12)). Two choices matter:

- ε **moves with** γ**.** It is rebuilt from (h, B, γ) at every point, never an input. Holding it fixed differentiates a different functional, and a check whose two sides share the omission cannot see it.
- **The orbitals are taken as they are.** ε is f's diagonal; off canonical orbitals nothing re-canonicalises. SC-NEVPT2 is not invariant to core–core or virtual–virtual rotations, so the pseudocanonical conditions $f_{ij} = f_{ab} = 0$ belong to the Lagrangian, not the functional.

Beyond the five MO blocks the builders read B_{cc} whole and B_{vv}'s diagonal.

6.2 The adjoints and the orbital source term

One reverse pass (§5, then the builders backwards, `rdms_adjoint` and the DF adjoints) gives

$$\bar{h} = \frac{\partial E_{\mathrm{corr}}}{\partial h}, \qquad \Lambda^L = \frac{\partial E_{\mathrm{corr}}}{\partial B^L}, \qquad y = \frac{\partial E_{\mathrm{corr}}}{\partial c}\Big|_{h,B}, \tag{20}$$

$\bar{h}$ and each Λ^L symmetrised last. Along $C \to C(1+\theta)$, any $\Phi(h, B)$ with symmetric adjoints moves by $\sum \theta_{xy} X_{xy}$,

$$X = 2\,h\,\bar{h} + 2\sum_L B^L \Lambda^L, \qquad A = X - X^{\mathsf{T}} = \frac{\partial \Phi}{\partial \kappa}, \tag{21}$$

evaluated as an AO half-transform ($\tilde{\Lambda}_q = \Lambda_q C^{\mathsf{T}}$ per q, one batched GEMM, then $D = B^{\mathsf{T}}_{\mathrm{AO}} \tilde{\Lambda}$, the DF-exchange shape) rather than pair by pair. Every block of A has a reader: the mixed blocks the Z-vector, core–core and virtual–virtual z^c, active–active the gauge identity. A is Park's orbital source term Y' [8].

6.3 The pseudocanonical multipliers

Along $\kappa_{pq} = -\kappa_{qp} = \theta$ within the core or the virtuals, f_{pq} and f_{qp} move by $\theta(\varepsilon_p - \varepsilon_q)$ and nothing else the constraint reads moves at first order. So stationarity of $E_{\mathrm{corr}} + \langle z^c, f \rangle$ (full Frobenius sum, symmetric z^c) decouples pair by pair:

$$z^c_{pq} = z^c_{qp} = -\frac{A_{pq}}{2(\varepsilon_p - \varepsilon_q)}, \qquad p \neq q \text{ both core or both virtual}, \tag{22}$$

Park's Eqs. 56–57 [8] and Eqs. 25–26 [10] in this tree's sign of κ. `zc_multipliers` takes only A and ε, so it serves SC, PC and QD alike.

Degenerate pairs. For $|\varepsilon_p - \varepsilon_q| < 10^{-6}\,E_h$, if $|A_{pq}| \leq 10^{-8}$ (partners in different irreps, say) $z^c_{pq} = 0$; otherwise the pair is refused, naming both orbitals. PC's A vanishes with the gap, so PC takes the zero branch where SC is refused: this is Park's reason for preferring PC [8].

The term z^c adds. $\Phi = \langle z^c, f \rangle$ with $D = 2$ on the core diagonal and γ (or γ^{SA}) on the active block, $g_L = \sum B^L_{rs} D_{rs}$ and $w_L = \sum B^L_{pq} z_{pq}$:

$$\Phi = \langle z, h \rangle + \sum_L w_L g_L - \tfrac{1}{2} \sum_L \mathrm{tr}(z B^L D B^L), \qquad \bar{h}^{\Phi} = z, \tag{23}$$

$$\Lambda^{\Phi, L} = z g_L + D w_L - \tfrac{1}{2}(z B^L D + D B^L z), \tag{24}$$

$$\mathcal{W} = \frac{\partial \Phi}{\partial \gamma} = \sum_L w_L B^L_{aa} - \tfrac{1}{2} \sum_L (B^L z B^L)_{aa}, \tag{25}$$

$$y^c_r = w_r \sum_{tu} (\mathcal{W} + \mathcal{W}^{\mathsf{T}})_{tu}\, E_{tu} c_r, \tag{26}$$

the last one sigma-vector application of a one-body operator, batched over the roots. y^c is raw (not projected), as y is.

6.4 Lagrangian, Z-vector, gate and nuclear term

On a converged single-state CASSCF,

$$\mathcal{L} = E_{\mathrm{CAS}} + E_{\mathrm{corr}} + Z \cdot g_{\mathrm{orb}} + z \cdot g_{\mathrm{ci}} + \langle z^c, f \rangle, \qquad g_{\mathrm{ci}} = 2(H_0 - E)c, \tag{27}$$

g_{orb} `response::fock`'s orbital gradient over the packed non-redundant pairs. Stationarity in the CASSCF parameters is the Z-vector equation [13] over the CI-relaxed Hessian, solved by `mcscf::zvector` (preconditioned CG, CI block projected off c): orbital right-hand side $g + \mathrm{pack}(A^{\mathrm{corr}} + A^{z^c})$ and CI right-hand side $\frac{1}{2}P_c(y + y^c)$. The $\frac{1}{2}$ is `zvector`'s convention (its CI row is half the Hessian's).

The gate. $\mathcal{L}$ is stationary in every rotation, so its total $X^{\mathcal{L}}$ must be symmetric in every block. Its antisymmetric part,

$$A^{\mathcal{L}} = 2(F - F^{\mathsf{T}})(\gamma + \gamma_z, \Gamma + \Gamma_z) + A^Z + A^{\mathrm{corr}} + A^{z^c}, \tag{28}$$

with F the generalised Fock at the root's RDMs plus z's transition RDMs, and $A^Z = 2(F' - F'^{\mathsf{T}})$, F' the derivative of F along $C(1 + \epsilon\kappa(Z))$, is evaluated and every block past 10^{-8} is refused, naming it. F is a polynomial of degree 4 in C, so a five-point stencil gives F' exactly to round-off. On the non-redundant pairs $A^{\mathcal{L}}$ is the Z-vector's residual, on the inactive pairs Eq. (22)'s, on the active pairs the gauge identity's: a wrong factor in any term names its block.

The nuclear term. Along any orthonormality-preserving path the symmetric part of θ is $-\frac{1}{2}C^{\mathsf{T}}S^A C$, so with the total adjoints $(\bar{h}^{\mathcal{L}}, \Lambda^{\mathcal{L}}, X^{\mathcal{L}})$,

$$\frac{dE}{dR_A} = \sum_{\mu\nu} \mathcal{P}_{\mu\nu} h^A_{\mu\nu} + \sum_{\mu\nu P} \mathcal{G}^P_{\mu\nu} (\mu\nu|P)^A - \tfrac{1}{2} \sum_{PQ} \mathcal{V}_{PQ} (P|Q)^A - \sum_{\mu\nu} \mathcal{W}_{\mu\nu} S^A_{\mu\nu} + V^A_{nn}, \tag{29}$$

$\mathcal{P} = C\bar{h}^{\mathcal{L}}C^{\mathsf{T}}$, $\mathcal{G}^P = \sum_L (L^{-\mathsf{T}})_{PL} C\Lambda^{\mathcal{L},L} C^{\mathsf{T}}$, $\mathcal{V} = L^{-\mathsf{T}}\mathcal{G}^{\mathrm{aux}}L^{-1}$ with $\mathcal{G}^{\mathrm{aux}}_{LL'} = \langle \Lambda^{\mathcal{L},L}, B^{L'} \rangle$, and $\mathcal{W} = \frac{1}{2}C\ \mathrm{sym}(X^{\mathcal{L}})C^{\mathsf{T}}$. In the code this is one `mcscf::df_gradient` pass: E_{CAS}'s and the multipliers' terms are the ones it builds, and $(\bar{h}, \Lambda, X)$ of $E_{\mathrm{corr}} + \Phi$ are its "extra" MO adjoint. No relaxed density is formed separately.

Algorithm 2 The DF-NEVPT2 nuclear gradient (`sc_gradient`, `pc_gradient`)

Input: converged single-state DF-CASSCF (C, c), core and virtuals canonical; SC or PC
Output: dE/dR

1: $(E_{\mathrm{corr}}, \bar{h}, \Lambda, y, X, A, \varepsilon) \leftarrow$ `sc_adjoint` or `pc_adjoint`
2: $z^c \leftarrow$ Eq. (22); refuse an undetermined degenerate pair
3: $(\Phi, \bar{h}^\Phi, \Lambda^\Phi, y^c, X^\Phi, A^\Phi) \leftarrow$ `zc_adjoint`; add field by field
4: $(Z, z) \leftarrow$ `zvector`$\big(g + \mathrm{pack}\,A,\ \frac{1}{2}(y + y^c)\big)$
5: **gate:** $A^{\mathcal{L}}$ below 10^{-8} in all six blocks
6: $dE/dR \leftarrow$ `df_gradient`$\big(\text{multipliers } (Z, z),\ \text{extra } (\bar{h}, \Lambda, X)\big)$

7 The QD-NEVPT2 gradient

7.1 The functional $\mathcal{E}[W]$

For any symmetric $n \times n$ W,

$$\mathcal{E}[W] = \sum_{MN} W_{MN} H^{\mathrm{eff}}_{MN} = \sum_M \rho_M E_M + \sum_{MN} W_{MN} C_{MN}, \qquad \rho_M = W_{MM}, \tag{30}$$

and state K's energy is $E_K = \mathcal{E}[u_K u_K^\mathsf{T}]$ ($u_K = R_K$; Park's R_{PPMN} [10]). R_K is an eigenvector of a symmetric matrix, so $\delta E_K = u_K^\mathsf{T} \delta H^{\text{eff}} u_K$: R needs no multiplier, and the gradient of E_K is that of $\mathcal{E}[W]$ at fixed W. Against a symmetric W the symmetrisation drops out, so the unsymmetrised couplings are what is differentiated. `qd_sc_energy` and `qd_pc_energy` build every input from (h, B, C) and the roots at their SA weights: one set of shared arrays, ε from f^{SA} with γ^{SA} rebuilt from the roots at every point, and each ordered pair's dm1–dm3.

The reverse pass. `qd_*_adjoint` runs the QD reverse pass at $\bar{H} = W$, hands each transition's dm1–dm3 bars (folded into dm3 along Eq. (12)) to the bra/ket `rdms_adjoint` with null f3 bars, then the shared builders backwards, so that $\bar{\gamma}^{\text{SA}}$ collects in one bar. Each root then takes $w_r \bar{\gamma}^{\text{SA}}$, whatever its W_{rr}: every root carries ε's share through γ^{SA}.

7.2 The Lagrangian on a state average

The SA-CASSCF parameters are the non-redundant rotations, the roots c_r and the rotations Θ among them. With (Z, z, Θ) in `mcscf::zvector`'s coordinates,

$$\mathcal{L} = \mathcal{E}[W] + Z \cdot g_{\text{SA}} + \sum_r 2w_r\, z_r \cdot (H - E_r)c_r + \sum_{r<s} \big(2(w_r - w_s)\Theta_{sr} + \zeta_{sr}\big)\langle c_s|H|c_r\rangle + \langle z^c, f^{\text{SA}}\rangle, \quad (31)$$

Park's Eq. 24 [10] made explicit about the rotations among the roots: a pair of unequal weights is a row of the SA Hessian (Θ), and a pair of equal weights, to which E_{SA} is blind, needs a multiplier ζ of its own. Unlike a CASSCF root energy, C_{MN} is not stationary under mixing the references, which is QD-NEVPT2's reference-basis dependence [7]. Along $c_r \to c_r + \theta c_s$, $c_s \to c_s - \theta c_r$,

$$\tau_{sr} = c_s^\mathsf{T}\bar{y}_r - c_r^\mathsf{T}\bar{y}_s, \qquad \zeta_{sr} = -\frac{\tau_{sr}}{E_s - E_r}\ (w_r = w_s), \qquad t_{sr} = \frac{\tau_{sr}}{2(w_r - w_s)}\ (w_r \neq w_s), \quad (32)$$

$\bar{y}_r$ the raw CI bar of $\mathcal{E}^{\text{corr}}[W] + \Phi^{\text{SA}}$. ζ's orbital term is $\zeta_{sr} g^{(sr)}$, the orbital gradient of $\langle c_s|H|c_r\rangle$ (`response::fock` at the transition RDMs less at zero RDMs), and its nuclear term is the Hellmann–Feynman gradient of $\langle c_s|H|c_r\rangle$, folded over all pairs into one set $(\gamma^\zeta, \Gamma^\zeta)$.

The Z-vector. Right-hand sides: orbital $g^\rho + \sum \zeta_{sr} g^{(sr)} + \text{pack}\, A$, g^ρ the Fock's orbital gradient at the ρ-weighted RDMs; CI $\bar{y}_r/(2w_r)$ for every root (`zvector`'s CI rows are the SA Hessian's divided by $2w_r$); rotations t. At $n = 1$ this is §6's $\frac{1}{2}(y + y^c)$. A root of weight below the rotation tolerance is refused, since it has no SA CI row. The gate is the single-state one at the ρ-weighted RDMs plus ζ's and z's.

Two derivative passes.

$$\frac{dE_K}{dR} = \underbrace{\texttt{df_gradient}\big(\rho;\ (Z, z, \Theta);\ \text{extra}\big)}_{\text{reference at } \rho,\ \text{multipliers, } \mathcal{E}^{\text{corr}}+\Phi^{\text{SA}}} + \underbrace{\texttt{df_transition_gradient}\big(\gamma^\zeta, \Gamma^\zeta\big)}_{\zeta,\ \text{skipped when } \zeta=0}. \quad (33)$$

7.3 The W form, and several W at once

Everything in $\mathcal{L}$ is linear in W except the core energy, which $\sum_M \rho_M E_M$ carries at $\text{tr}\, W$. The W form (`qd_*_gradient(..., w)`) takes any symmetric W: $\rho = \text{diag}\, W$; g^ρ and the gate carry the core part $\text{tr}\, W$ times; and the first pass is one `df_gradient` at the mix ρ when $\sum \rho = 0$ (its core-off path) or $\sum \rho = 1$ with $\rho \geq 0$. Any other $\rho = (\text{tr}\, W)p + d$, $p \geq 0$ normalised, takes one

unrelaxed pass at p scaled by $\operatorname{tr} W$ and one relaxed pass at d. So $G(W_1 + W_2) = G(W_1) + G(W_2)$ and $\sum_K dE_K/dR = G(1)$. The state form is a wrapper that builds $W = u_K u_K^\mathsf{T}$, refusing a QD state within $10^{-8}\,E_h$ of another.

A list of W's shares the forward arrays, the Fock, the CI-relaxed Hessian and one block Z-vector solve (`mcscf::zvectors`) over all right-hand sides, each column's true residual checked against 10^{-10}. The derivative passes stay one per W: one three-centre derivative kernel is most of each pass, so stacking them would need a multi-density kernel.

Algorithm 3 The DF-QD-NEVPT2 gradient $G(W)$ (`qd_lagrangians` and `qd_passes`)

Input: DF-SA-CASSCF $(C, \{c_r\}, w, \{E_r\})$, core and virtuals canonical over f^{SA}; SC or PC; symmetric W (state form: $W = u_K u_K^\mathsf{T}$ from H^{eff})

Output: $G(W)$; for the state form dE_K/dR

1: $(\bar{h}, \Lambda, \bar{y}_r, X, A, \varepsilon) \leftarrow$ `qd_sc_adjoint` or `qd_pc_adjoint` at W
2: $z^c \leftarrow$ Eq. (22) over A and $\varepsilon^{\mathrm{SA}}$; Φ^{SA}'s adjoints at the weights; add
3: τ; ζ on equal-weight pairs, t on unequal ones; ζ's orbital gradient and $(\gamma^\zeta, \Gamma^\zeta)$
4: $\rho \leftarrow \operatorname{diag} W$; g^ρ at the ρ-weighted RDMs, core part at $\operatorname{tr} W$
5: $(Z, z, \Theta) \leftarrow$ `zvector`$(g^\rho + \zeta \cdot g + \operatorname{pack} A,\ \bar{y}_r/(2w_r),\ t)$
6: **gate** per block
7: $G \leftarrow$ `df_gradient`$(\rho, (Z, z, \Theta), \mathrm{extra})$ + `df_transition_gradient`$(\gamma^\zeta, \Gamma^\zeta)$

The dropped rank-projector terms (PC). Eq. (16) and its QD form differentiate the restricted inverse as if S_N's kept span were fixed, exact when S_N's dropped modes annihilate $|N\rangle$, at eigenstates. Ladders along each real root c_r are what check it, since random RDM directions carry no eigenstate structure. The residual terms of an unconverged CASCI are the ones Kollmar et al. [12] show reach the 5-RDM in the Dyall matrix elements.

8 Interstate and derivative couplings

8.1 The interstate coupling

With $W^{PQ} = \frac{1}{2}(u_P u_Q^\mathsf{T} + u_Q u_P^\mathsf{T})$ (Park's R_{PQMN}),

$$\mathbf{h}_{PQ} = u_P^\mathsf{T}(\nabla H^{\mathrm{eff}})u_Q = G(W^{PQ}), \qquad \operatorname{tr} W^{PQ} = 0, \tag{34}$$

so the first pass takes `df_gradient`'s core-off path: the reference part $\sum_r R_{rP} R_{rQ} \nabla E_r$ has no core energy and no nuclear repulsion. Since $u_P^\mathsf{T} \delta H^{\mathrm{eff}} u_Q = (E_Q - E_P) u_P^\mathsf{T} \delta u_Q$,

$$\mathbf{d}_{PQ}^{\mathrm{NEV}} = \sum_M R_{MP} \nabla R_{MQ} = \frac{\mathbf{h}_{PQ}}{E_Q - E_P}. \tag{35}$$

A pair within $10^{-8}\,E_h$ is refused. $\mathbf{h}_{PQ}$ carries R_P's sign times R_Q's, and so the roots' phases only through them.

8.2 The derivative coupling

With $\tilde{\Psi}_P = \sum_M (|M\rangle + |\Psi_M^{(1)}\rangle) R_{MP}$, Park's Eqs. 45–50 [10] (after the MS-CASPT2 derivation of Park and Shiozaki [11]) give

$$\mathbf{d}_{PQ} = \mathbf{d}_{PQ}^{\mathrm{NEV}} + \mathbf{d}_{PQ}^{\mathrm{CAS}} + \mathbf{d}_{PQ}^{\mathrm{PT2}}, \qquad T_{MN}^{PQ} = \tfrac{1}{2}(R_{MP}R_{NQ} - R_{MQ}R_{NP}), \tag{36}$$

$$\mathbf{d}_{PQ}^{\mathrm{CAS}} = \sum_{MN} T_{MN}^{PQ} \langle M|\nabla N\rangle, \qquad \mathbf{d}_{PQ}^{\mathrm{PT2}} = \sum_{MN} T_{MN}^{PQ} \langle \Psi_M^{(1)}|\nabla N\rangle = \sum_{xy} D_{xy}^{\mathrm{PT2}} \langle \phi_x|\nabla \phi_y\rangle, \tag{37}$$

the last because $\Psi_M^{(1)}$ lies outside the active space, so ∇ reaches it only through $|N\rangle$'s orbitals, with

$$D_{xy}^{\mathrm{PT2}} = \sum_{MN} T_{MN} \langle \Psi_M^{(1)}|E_{xy}|N\rangle. \tag{38}$$

Both $\mathbf{d}^{\mathrm{CAS}}$ and $\mathbf{d}^{\mathrm{PT2}}$ are one `mcscf::df_nac` combination

$$\mathbf{d}^T = \sum_{MN} T_{MN}\mathbf{d}_{MN} + \mathbf{d}[D^X], \qquad \mathbf{d}[D^X] = \sum_{xy} D_{xy}^X \langle \phi_x|\nabla\phi_y\rangle, \tag{39}$$

with the SA orbitals and CI relaxed: $\mathbf{d}^{\mathrm{CAS}}$ at $(T^{PQ}, D^X = 0)$ and $\mathbf{d}^{\mathrm{PT2}}$ at $(T = 0, D^X = D^{\mathrm{PT2}})$. Its pieces, by linearity in the pair coupling:

- **CI part:** the transition Lagrangian at the combined RDMs $\sum_{M<N} 2T_{MN}/(E_N - E_M)\,(\gamma^{(MN)}, \Gamma^{(MN)})$ (symmetrised), one Z-vector and one `df_transition_gradient`.
- **CSF part:** $\frac{1}{2}G[D^{\mathsf{T}} - D]$ at $D = C_a \gamma^T C_a^{\mathsf{T}}$, $\gamma^T = \sum_{MN} T_{MN}\gamma^{(MN)}$, with $G[a]_{A,k} = \sum_{m\in A}\sum_n a_{mn}\,\partial S_{mn}/\partial A_k$ the half-derivative overlap (PySCF's `nac_csf`).
- **D^X:** with $C(R) = C_0(C_0^{\mathsf{T}} S(R) C_0)^{-1/2} e^{\kappa}$, the *basis motion* $\frac{1}{2}G[C(D^{X\mathsf{T}} - D^X)C^{\mathsf{T}}]$, and the *orbital response*, $D_{pq}^X - D_{qp}^X$ added to the Z-vector's orbital right-hand side. D^X has no CI part.

`mult_ediff`	`use_etfs`	`total`
false	false	$\mathbf{h}_{PQ}/\Delta E + \mathbf{d}^{\mathrm{CAS}} + \mathbf{d}^{\mathrm{PT2}}$ (default)
false	true	$\mathbf{h}_{PQ}/\Delta E + \mathbf{d}^{\mathrm{CAS}} + \mathbf{d}^{\mathrm{PT2}}$ − csf
true	false	$\mathbf{h}_{PQ} + \Delta E(\mathbf{d}^{\mathrm{CAS}} + \mathbf{d}^{\mathrm{PT2}})$
true	true	$\mathbf{h}_{PQ} + \Delta E(\mathbf{d}^{\mathrm{CAS}} + \mathbf{d}^{\mathrm{PT2}} - \mathrm{csf})$

$\Delta E = E_Q - E_P$, and csf is $\mathbf{d}^{\mathrm{CAS}}$'s CSF part plus $\mathbf{d}^{\mathrm{PT2}}$'s basis motion (the terms electron-translation factors drop).

8.3 The PT2 transition density

T is antisymmetric, so only $M \neq N$ enter and every term that reads no RDM carries $\langle M|N\rangle = 0$. $E_{xy}|N\rangle$ lies in the first-order interacting space only for $(x, y) = (t, i)$, (a, t) and (a, i), one class each (Si, Sr, Sir); every other element of D^{PT2} is exactly zero. In that class $E_{xy}|N\rangle$ is the one-body function χ_y^N of V's h_v term, and $\Psi_M^{(1)}$ is a combination of the class's internal functions ϕ_μ^M (and, for SC, of χ^M). So with the T-weighted transition RDMs $\rho^M = \sum_N T_{MN}\gamma^{(MN)}$ (dm1 and dm2, bra M), each block needs only

$$\langle \phi_\mu^M|E_{xy}|\rho^M\rangle = S_1(\rho^M)[\mu, y], \qquad \langle \chi_a^M|E_{xy}|\rho^M\rangle = S_{11}(\rho^M)[a, y], \tag{40}$$

S_1 the class's PC one-body metric and S_{11} SC's $h_v \cdot h_v$ metric, both over ρ^M. No dm3, no f3, no 4-RDM.

class	block	$S_1(\rho)$	$S_{11}(\rho)$
Sr, over r	$D[r,u]$	$\rho_2[r',p,q,u]$, $\mu=(p,q,r')$	$\rho_1[a,u]$
Si, over i	$D[u,i]$	$\mathrm{dm2_h}(\rho)[r,p,q,u]$	$-\rho_1[u,a]$
Sir, over (r,i)	$D[r,i]$	$[2\rho_1[q,p];\,-\rho_1[q,p]]$, $\mu=(x,p,q)$	0

with $\mathrm{dm2_h}(\rho)[a,b,c,d] = 2\rho_1[a,b]\delta_{cd} - \rho_2[a,b,d,c]$.

SC. With the handed-over c_t^M and slab rows x,

$$D[r,u] = \sum_M c_r^M \Big(\sum_{pqr'} x[r,p,q,r']\,\rho_2^M[r',p,q,u] + \sum_a (h_v^{\mathrm{Sr}})_{ra}\,\rho_1^M[a,u] \Big), \tag{41}$$

$$D[u,i] = \sum_M c_i^M \Big(\sum_{pqr} x[q,p,i,r]\,\mathrm{dm2_h}(\rho^M)[r,p,q,u] - \sum_a (h_v^{\mathrm{Si}})_{ai}\,\rho_1^M[u,a] \Big), \tag{42}$$

$$D[r,i] = \sum_M c_{ri}^M \sum_{pq} \big(2x_1[r,p,i,q] - x_2[r,p,q,i]\big)\rho_1^M[q,p], \tag{43}$$

one more walk of the Sr, Si, Sir1 and Sir2 slabs.

PC. $\Psi_M^{(1)} \ni -\sum_k g_{tk}^M \sum_\mu U_M[\mu,k]\phi_\mu$ has no χ part, so with $W^M = S_1(\rho^M)^{\mathsf{T}} U_M$, $D[\cdot,t] = -\sum_M \sum_k W^M[\cdot,k]\, g_{tk}^M$: two einsums per state and class, no slab.

8.4 Three solves or one: the fold

The default path solves three Z-vectors per pair ($\mathbf{h}_{PQ}$'s, and the combination twice) so that $\mathbf{h}_{PQ}$, $\mathbf{d}^{\mathrm{CAS}}$ and $\mathbf{d}^{\mathrm{PT2}}$ can be reported apart. Every right-hand side is linear in its source and all three share the CI-relaxed SA Hessian, so Park's Eqs. 51–52 [10] fold them:

$$\mathbf{h}_{PQ} + \Delta E(\mathbf{d}^{\mathrm{CAS}} + \mathbf{d}^{\mathrm{PT2}} - \mathrm{csf}) = G(W^{PQ})\Big|_{\substack{\mathrm{RDMs}+\Delta E(\gamma,\Gamma)_{T^{PQ}} \\ A+\Delta E(D^{\mathrm{PT2}}-D^{\mathrm{PT2T}})}}, \tag{44}$$

one Z-vector whose orbital right-hand side and Hellmann–Feynman pass take ΔE times the combination's RDMs, and whose A takes ΔE times D^{PT2}'s antisymmetric part; csf needs no Z-vector. The fold's RDMs are no state's, so their own active–active asymmetry (the CSF term's, which no multiplier removes) is taken out of the gate in that block. It costs about 0.55–0.68 of the three-solve path, and is off by default because it cannot report the three parts separately.

Conventions. Bra P, ket Q: $\mathbf{d}_{PQ} = -\mathbf{d}_{QP}$, $\mathbf{h}_{PQ} = \mathbf{h}_{QP}$ (PySCF's `kernel(state=(I,J))` is $\langle I|\partial J\rangle$). $\mathbf{d}_{PQ}$ changes sign with the CI phases as R carries them, by $s_{m(P)}s_{m(Q)}$, $m(P)$ the index of R_P's largest component; no phase is fixed by the call.

9 Conical intersections

`qd_*_intersection` returns $\mathbf{g}_P = G(u_P u_P^{\mathsf{T}})$, $\mathbf{g}_Q = G(u_Q u_Q^{\mathsf{T}})$ and $\mathbf{h}_{PQ} = G(W^{PQ})$ from one set of H^{eff} eigenvectors, by default through one several-W call (one shared block Z-vector). It does not refuse a degenerate pair: G is linear in W and nothing divides by $E_Q - E_P$. At degeneracy u_P, u_Q are any orthonormal pair of the two-state space and the three vectors turn with them, but the branching plane $\mathrm{span}\{\mathbf{g}_Q - \mathbf{g}_P, \mathbf{h}_{PQ}\}$ and $\frac{1}{2}(\mathbf{g}_P + \mathbf{g}_Q)$ do not, which is all `geometry::optimize_meci` uses.

Algorithm 4 The QD-NEVPT2 derivative coupling (`qd_sc_nac`, `qd_pc_nac`)

Input: DF-SA-CASSCF; SC or PC; QD states $P \neq Q$; `NacOptions`; fold
Output: $\mathbf{d}_{PQ}$ and its parts

H^{eff}, R (signed), $\Delta E = E_Q - E_P$; refuse $|\Delta E| < 10^{-8}$
$T^{PQ} \leftarrow \frac{1}{2}(R_P R_Q^{\mathsf{T}} - R_Q R_P^{\mathsf{T}})$ ▷ exactly antisymmetric
$D^{\text{PT2}} \leftarrow$ Eq. (38) at T^{PQ} over the handed-over amplitudes
if fold **then**
    $(\gamma, \Gamma, \text{csf}) \leftarrow$ `nac_combination_terms`(T^{PQ}, D^{PT2})
    $F \leftarrow G(W^{PQ})$ with $\Delta E(\gamma, \Gamma)$ and $\Delta E(D^{\text{PT2}} - D^{\text{PT2}\mathsf{T}})$ folded in
    `total` $\leftarrow F/\Delta E +$ csf (per the options)
else
    $\mathbf{h}_{PQ} \leftarrow G(W^{PQ})$
    $\mathbf{d}^{\text{CAS}} \leftarrow$ `df_nac`(T^{PQ}); $\mathbf{d}^{\text{PT2}} \leftarrow$ `df_nac`$(0, D^{\text{PT2}})$
    `total` $\leftarrow \mathbf{h}_{PQ}/\Delta E + \mathbf{d}^{\text{CAS}} + \mathbf{d}^{\text{PT2}}$ (per the options)

Gap sensitivities. An element $a_W = \text{tr}(WH^{\text{eff}})$ is not stationary in the SA-CASSCF parameters, so a solution with residual $\mathcal{G}$ moves it by $\delta a_W = -z^W \cdot \mathcal{G}$ to first order, z^W the W-form's Z-vector. In the pair's 2×2 block $[[a, b], [b, d]]$ the gap is $\sqrt{(a-d)^2 + 4b^2}$, and

$$|\delta \,\text{gap}| \leq s_o \|g\| + s_c \sqrt{n}\, \rho_{\text{CI}}, \qquad s_o = \sqrt{\|z_o^Q - z_o^P\|^2 + 4\|z_o^{PQ}\|^2}, \quad s_c = \sqrt{\|\hat{z}_c^Q - \hat{z}_c^P\|^2 + 4\|\hat{z}_c^{PQ}\|^2}, \tag{45}$$

$\hat{z}_{c,r} = 2w_r z_{c,r}$, $\|g\|$ the SA orbital gradient norm and ρ_{CI} the largest Davidson residual. Θ needs no term (Rayleigh–Ritz) and z^c none (a direct diagonalisation). Both s are invariant to turning u_P, u_Q within their pair.

10 Numerical hazards and refusals

- **SC's inactive non-invariance.** SC is not invariant to rotations among inactive orbitals, so z^c divides by orbital-energy gaps (Eq. (22)); a degenerate pair with a finite source is refused. PC takes the zero branch there. Park finds SC's gradients unstable for this reason and recommends PC [8].
- **SC's small-norm pole.** A perturber whose own norm is tiny (around 10^{-8} on some symmetric cases) has a Rayleigh quotient H/N that round-off moves, and $c_t = -1/(\Delta + H/N)$ amplifies it by $\sim 1/N$. The energy cannot see it (cN is small); the bars and the PT2 density can. PC divides by no norm and is unaffected.
- **PC's metric cut.** The gap rule refuses an ambiguous rank; a non-positive $\lambda + \Delta$ is refused. Near either, the energy is not smooth, and the derivatives inherit the solve's conditioning.
- **Gaps.** QD states closer than $10^{-8}\,E_h$ are refused by the state form, the coupling and the NAC (not by the W forms or the intersection); CASSCF roots closer than that which ζ, t or T^{PQ} couple are refused.
- **Reproducibility.** The einsums and the DF derivative kernels sum with atomics, so two runs differ at the 10^{-12} relative level (more where a solve is ill-conditioned); `rdms` and its adjoint are bitwise reproducible.

- **Convergence.** Every derivative assumes a converged SA-CASSCF; the gate measures how far the Lagrangian is from stationary, and the gap sensitivities bound the effect on a QD gap.

11 Verification

Each acceptance bar is fixed by analysis and checked once: 10^{-10} for two paths to the same number, 10^{-7} relative for a Richardson ladder of central differences (successive changes falling by a factor in $(3.2, 4.8)$ over three halvings), and twice the committed error bar for a numerical reference.

quantity	checked against
SC energies	PySCF DF-NEVPT2, every class, $10^{-7}\,E_h$
PC energies	block2 `WickICNEVPT2`, every class and dropped count
SC-, PC-QD H^{eff}	a brute-force reference: every element's per-class coupling, 10^{-10}
dm3, f3ca, f3ac	PySCF, 10^{-12} of max; tiled vs one tile, 10^{-13}
RDM adjoints	polarisation identities (the build is quadratic in c, linear in $(tu\|vw)$)
class reverse passes	Eq. (15), homogeneity and Euler's degree 4, 10^{-12}; ladders along B, h_v, ε, h, RDMs
$(\bar{h}, \Lambda, y, A)$	ladders along h, B, each rotation block and c, every input rebuilt
z^c	stationarity of $E_{\text{corr}} + \Phi$ along core–core and virtual–virtual rotations
Lagrangians	stationarity ladders (orbitals, CI off every root, rotations among roots) at equal, unequal and mixed weights; they pin $1/(2w_r)$
gradients	numerical gradients, $2\times$ bar; translation and rotation sums
W form	$G(W_1 + W_2) = G(W_1) + G(W_2)$, 10^{-10}
$\mathbf{h}_{PQ}$, $\mathbf{d}^{\text{NEV}} + \mathbf{d}^{\text{CAS}}$	numerical references, $2\times$ bar; $\mathbf{h}_{QP} = \mathbf{h}_{PQ}$, $\mathbf{d}_{QP} = -\mathbf{d}_{PQ}$
$\mathbf{d}_{PQ}$, $\mathbf{d}^{\text{PT2}}$, D^{PT2}	a brute-force reference, $2\times$ bar; 10^{-10} of max $\lvert D\rvert$
fold	the three-solve path, 10^{-9} (two independently converged solves)
intersection	each W form at the same u's; the gap bound on a stopped SA-CASSCF

12 Map to the code and to the papers

here	code
§4.1	`nevpt2::mo_blocks`, `h2e`, `slab`, and `*_adjoint`
§4.2	`nevpt2::rdms`, `rdms_adjoint` (one vector; bra/ket)
Eqs. (4), (5)	`class_energies`, `pc_class_energies`
Eqs. (6)–(7)	`sc_effective_hamiltonian`, `pc_effective_hamiltonian`
§5	`class_energies_adjoint`, `pc_class_energies_adjoint`, `*_effective_hamiltonian_adjoint`
§6	`sc_energy`, `sc_adjoint`, `pc_*`, `zc_multipliers`, `zc_adjoint`, `lagrangian_gradient`, `sc_gradient`, `pc_gradient`
§7	`qd_*_energy`, `qd_*_adjoint`, `qd_sc_lagrangian`, `qd_*_gradient` (state, W, several W)
§8	`qd_*_coupling`, `*_pt2_transition_density`, `qd_*_nac`
§9	`qd_*_intersection`

here	Park
Eq. (14), Eq. (15)	[8], Eqs. 36–38, 40
Eq. (22)	[8], Eqs. 56–57; [10], Eqs. 25–27
Eq. (27)	[8], Eq. 23
Eq. (8), Eq. (30)	[10], Eqs. 16–20 ($R_{PPMN} = W_{MN}$)
the QD reverse passes	[10], Eqs. 21–23 (λ, implicit in the bars), 31–43
Eq. (31)	[10], Eq. 24, with ζ and Θ explicit
the Z-vector right-hand sides	[10], Eqs. 28–29
§8	[10], Eqs. 45–54

Where this departs from Park.

- Park's QD gradient is PC only, with SC called straightforward [10]; both are implemented here, with SC's hazards stated.
- Park contracts 4-RDM derivatives through multipass intermediates ([10], Eqs. 55–58). Here the f3ca/f3ac build is differentiated directly, tile by tile, so neither the 4-RDM nor its derivative exists. Kollmar et al.'s factorisation of the Dyall matrix elements [12] would remove f3 altogether; it is not used.
- ε is a function of γ (or γ^{SA}) inside the functional, the multipliers among the roots (ζ, Θ) are explicit, and the reference block is the diagonal form $E_M\delta_{MN}$, equal to Park's on the constraint surface.
- Printed misprints affect a reader of [10]: Eq. 5 drops the two-electron integral of the active term; Eq. 16 omits $\langle M|H|N\rangle$; Eq. 29's $z^c_{xy}\langle I|E_{xy}|N\rangle$ is zero as printed and should be $[\mathbf{g}(z^c)]_{xy}$ ($\mathcal{W}$ here); Eq. 41 has $\Gamma^{(3)}_{a'a,b'b,ec}$ where Eq. 39 has $\Gamma^{(3)}_{a'd,b'b,ec}$; Eq. 56 sums over its free indices.

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